

Testing gRPC Services Using BloomRPC



Google introduced the gRPC framework in 2015. As the name suggests, this framework is based on the implementation of RPC (remote procedure call). Developed on top of Http/2, it is used for applications with low latency and high throughput transactions. This article focuses on how BloomRPC is used for testing gRPC services.

gRPC is developed on Http/2 and follows the client-response communication model. These attributes enable it to support bi-directional and streaming communication, as it has the capability to receive a request from multiple clients and respond to them simultaneously by constantly streaming information.

gRPC supports multiple unary and streaming interactions:

- Unary
- Server streaming
- Client streaming
- Bi-directional streaming

It has limited browser support and requires gRPC-web and a proxy layer to perform conversions between HTTP 1.1 and HTTP 2.

Documentation of services

Service definition and payload information are described as part of *protobuf* files. Unlike in Rest, we don't need Swagger export to look into it. A *protobuf* file describes the interfaces of services with their types. A sample of a *protobuf* file is

given below:

```
syntax = "proto3";

enum BookCategory {
  MYSTERY = 0;
  SCIENCE FICTION = 1;
  SELF HELP = 2;
}

message RecommendationRequest {
  int32 user_id=1;
  BookCategory category = 2;
  int32 max_results = 3;
}

message BookRecommendation {
  int32 id= 1;
  string title 2;
}

message RecommendationResponse {
  repeated BookRecommendation
  recommendations = 1;
}

service Recommendations {
  rpc Recommend (RecommendationRequest)
  returns (RecommendationResponse);
}
```

Testing gRPC

Various tools are available to carry out

gRPC testing. These include grpc-curl, grpcui, Postman2, grpc-proxy, grpc-dump, Krey, Insomnia, Wombat, and BloomRPC.

Each of these tools tests and debugs grpc calls in unique ways. Tools like grpc-proxy help to intercept calls, just like Fiddler.

In this article, we will learn about how BloomRPC, a widely used gRPC client, is used for testing.

BloomRPC

BloomRPC is an open source project with an LGPL-3.0 licence. It supports Windows, Mac, and different flavours of Linux OSes.

Setup: You can install BloomRPC by using its installer, which can be found at <https://github.com/bloomrpc/bloomrpc/releases>.

Package managers can also be used for the installation of BloomRPC.

macOS: The Homebrew package manager is used for the macOS:

```
brew install --cask bloomrpc
```

Windows: On Windows, you can use the Chocolatey package manager:

```
choco install bloomrpc
```

Overview of the tool

Importing the *protobuf* path: It's important to add a path, to enable all the sub-folders and files that are referenced as the *path* of the *protobuf* definition to be considered.

You can add the *protobuf* path, as shown in Figure 1.

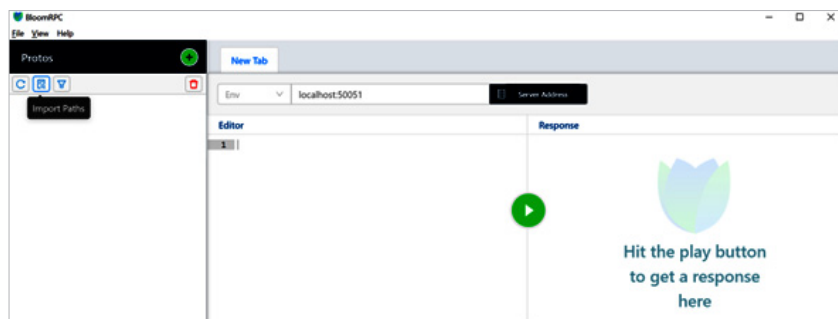


Figure 1: Import *path*